Supplementary Material for "SGF-RWKV: Semantic-Guided and Frequency-Enhanced RWKV for Unified Medical Image Denoising"

S1. Detailed Unified Training Protocol

To improve reproducibility, we provide the complete training protocols of SGF-RWKV in Table S1. For the task-specific comparison in Table I, SGF-RWKV is independently trained on each individual dataset following the conventional single-task setting. For all remaining experiments, including unified denoising comparison (Table II), ablation studies, zero-shot evaluation, and downstream segmentation analysis, a single SGF-RWKV model is jointly trained on all five datasets using dataset-balanced sampling. At each iteration, one dataset is randomly selected with equal probability, followed by random sampling from the selected dataset. This strategy prevents datasets with larger numbers of training samples from dominating the optimization process. Each sample is paired with its corresponding textual degradation description, while all trainable network parameters are jointly optimized across datasets.

Table S1: Unified training protocol summary.

Item	**Configuration**
Training setting	Single-task training for Table I; unified joint training for all other experiments
Datasets	OCT-Sp, US-Sp, CryoEM-N, MRI-N, LoDoPaB-CT
Batch sampling	Dataset-balanced sampling (uniform probability over datasets)
Prompt input	Predefined textual degradation descriptions according to degradation types
Shared parameters	All trainable network parameters are jointly optimized
Text encoder	Frozen CLIP text encoder for semantic prompt encoding
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	2×10^{-4} to 1×10^{-6} (cosine annealing)
Batch size	2
Total iterations	300,000
Training time	Approximately 158 h on a single NVIDIA A6000 GPU

S2. Statistical Significance Analysis

To verify whether the observed performance improvements over the strongest baseline Restore-RWKV are statistically significant, we perform paired two-sided Wilcoxon signed-rank tests on per-image PSNR, SSIM, and RMSE differences across all five test sets. The detailed results are reported in Table S2. For each dataset, $\Delta\mu$ denotes the average performance gap between SGF-RWKV and Restore-RWKV, σ represents the standard deviation of image-level performance differences, and the p-value evaluates whether the observed improvement is statistically significant.

All five datasets achieve statistically significant improvements with $p < 10^{-6}$ for all three evaluated metrics, including those with relatively small average gains such as CryoEM-N (+0.14 dB PSNR) and OCT-Sp (+0.30 dB PSNR). All experiments were conducted under the same controlled setting with a fixed random seed (seed = 42) to ensure reproducibility.

Table S2: Wilcoxon signed-rank test: SGF-RWKV vs. Restore-RWKV.

	PSNR			SSIM			RMSE		
Dataset	$\Delta\mu$	σ	p	$\Delta\mu$	σ	p	$\Delta\mu$	σ	p
OCT-Sp	+0.30	1.10	$<10^{-6}$	+0.0034	0.0042	$<10^{-6}$	−0.0012	0.0034	$<10^{-6}$
US-Sp	+1.72	2.30	$<10^{-6}$	+0.0139	0.0156	$<10^{-6}$	−0.0116	0.0065	$<10^{-6}$
CryoEM-N	+0.14	0.48	$<10^{-6}$	+0.0027	0.0085	$<10^{-6}$	−0.0005	0.0016	$<10^{-6}$
LoDoPaB-CT	+0.41	1.44	$<10^{-6}$	+0.0048	0.0175	$<10^{-6}$	−0.0031	0.0041	$<10^{-6}$
MRI-N	+0.29	0.75	$<10^{-6}$	+0.0075	0.0038	$<10^{-6}$	−0.0014	0.0022	$<10^{-6}$

Two-sided Wilcoxon signed-rank test. All evaluated metrics achieve $p < 10^{-6}$.
Positive $\Delta\mu$ indicates improvement for PSNR/SSIM, while negative $\Delta\mu$ indicates improvement for RMSE.
Test sample sizes: OCT-Sp (359), US-Sp (255), CryoEM-N (351), LoDoPaB-CT (356), MRI-N (317).

S3. Cosine Distance Among CLIP Prompt Embeddings

To characterize the structure of the CLIP text embedding space, we compute pairwise cosine distances $(1 - \cos(\mathbf{e}_i, \mathbf{e}_j))$ among the five prompt embeddings after L2 normalization. Cosine distance is the natural metric for CLIP representations: CLIP was trained with a contrastive objective over normalized embeddings, and SGM's internal attention mechanism likewise operates via dot-product similarity in the same normalized space. The distance matrix is reported in Table S3.

The distance matrix reveals a graded structure. The closest pair is "Gaussian noise in MRI" and "noise in low dose CT" (0.0525), followed by "noise in cryo-EM image" and "noise in low dose CT" (0.0567)—both sharing the phrase "noise in." The two speckle-related prompts exhibit a cosine distance of 0.0631, while unrelated pairs such as "speckle in OCT" and "noise in cryo-EM image" are farthest apart (0.1089). This structured embedding space provides a quantitative basis for predicting transfer behaviour: if CLIP conditioning is sensitive to embedding-space geometry, prompts closer to the correct one should better preserve restoration performance.

Table S3: Pairwise cosine distances between CLIP prompt embeddings.

	SO	**SU**	**NC**	**NL**	**GM**
SO	–	0.0631	0.1089	0.0706	0.0796
SU		–	0.0611	0.0665	0.0760
NC			–	0.0567	0.0722
NL				–	0.0525
GM					–

SO: speckle in OCT; SU: speckle in ultrasound; NC: noise in cryo-EM image; NL: noise in low dose CT; GM: Gaussian noise in MRI. Cosine distance is defined as $1 - \cos(\mathbf{e}_i, \mathbf{e}_j)$ over L2-normalized CLIP ViT-B/32 text embeddings. Smaller values indicate higher semantic similarity.

S4. Training Convergence Analysis

To evaluate the optimization stability of the proposed unified framework, we record the training loss and validation PSNR throughout the entire training process. As shown in Fig. S4, the training loss exhibits a decreasing trend with expected fluctuations, while the validation PSNR gradually improves and approaches a stable plateau toward the end of training. This indicates stable optimization behaviour throughout the 300,000-iteration training process.

S5. SGM Score Maps Under Unseen Prompt Transfer

To qualitatively examine how SGM responds to different text prompts, we visualize the multi-scale score maps generated under correct and unseen prompt conditions. All score maps are normalized using the same

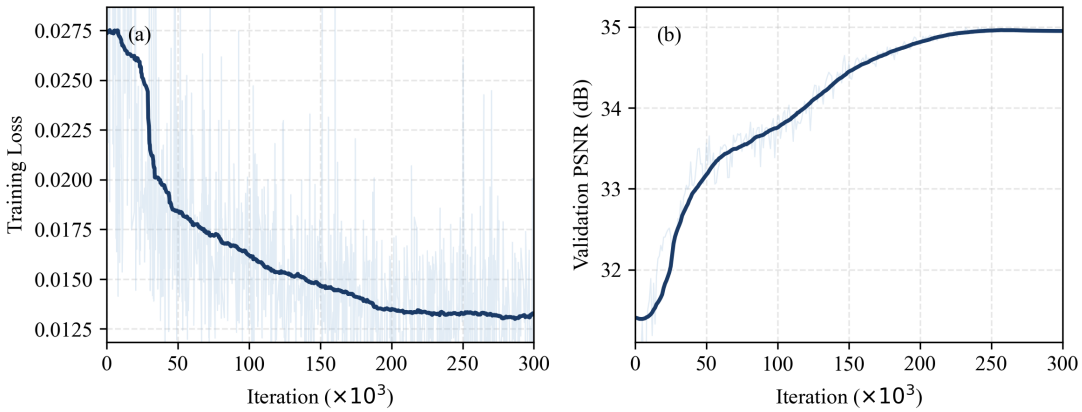


Figure S4: Training convergence curves of SGF-RWKV over 300,000 iterations. (a) Training loss consisting of reconstruction loss and $\mathcal{L}_{\mathrm{SM}}$. (b) Validation PSNR averaged over the five validation datasets. The model exhibits stable optimization behaviour, and the validation PSNR gradually reaches a plateau after approximately 250,000 iterations.

global range across different prompts for fair comparison.

As shown in Fig. S5, prompts sharing the same degradation type (e.g., "speckle in ultrasound" applied to OCT images) produce similar spatial activation patterns to the correct prompt, mainly concentrating on degradation-related texture regions. In contrast, prompts describing different degradation types (e.g., "noise in low-dose CT" applied to OCT images) generate distinct spatial responses. These observations indicate that changing the prompt modifies the spatial modulation process of SGM according to the degradation description, rather than simply selecting a fixed pre-learned task embedding.

S6. SGM Score Maps With and Without $\mathcal{L}_{\mathrm{SM}}$

To illustrate the effect of the $\mathcal{L}_{\mathrm{SM}}$ auxiliary loss on the spatial guidance mechanism, we compare SGM score maps generated by models trained with and without $\mathcal{L}_{\mathrm{SM}}$. Sample visualizations for OCT and CT modalities are shown in Fig. S6.

The score maps trained with $\mathcal{L}_{\mathrm{SM}}$ tend to produce more spatially structured activation patterns concentrated on degradation-relevant regions, whereas score maps trained without $\mathcal{L}_{\mathrm{SM}}$ exhibit more diffuse responses. These qualitative observations are consistent with the quantitative ablation results, supporting the interpretation that $\mathcal{L}_{\mathrm{SM}}$ acts as an auxiliary regularizer that encourages the semantic-guided score maps to focus on degradation-relevant spatial structures rather than simply tracking high-frequency artifacts.

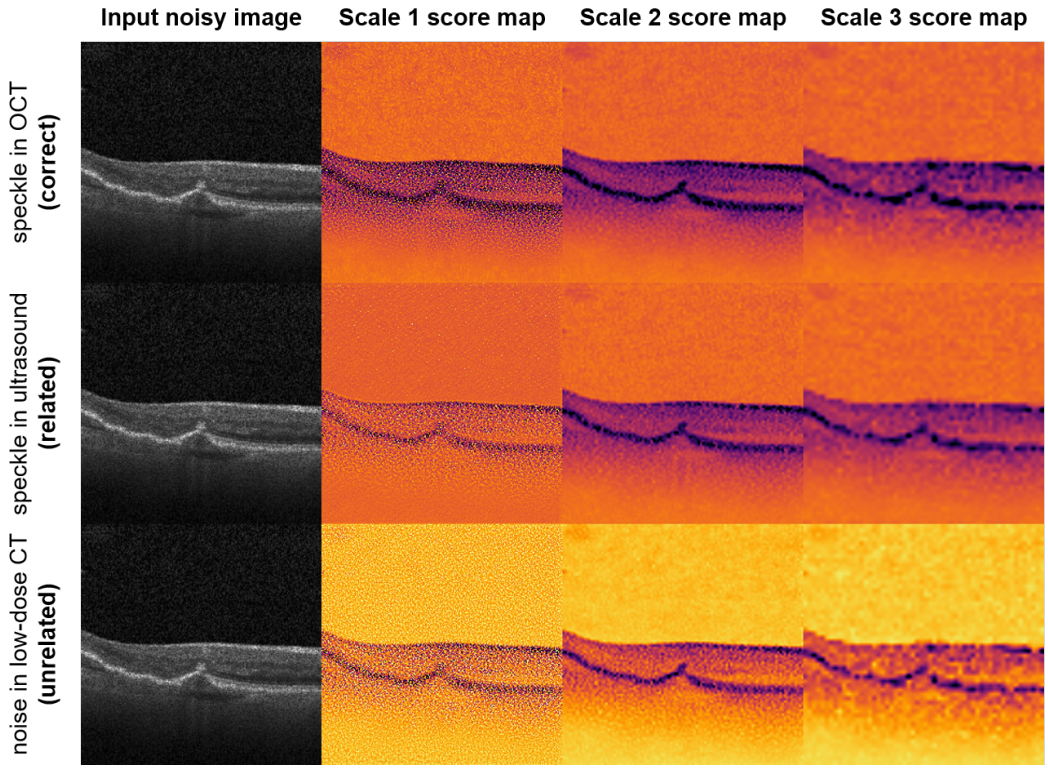


Figure S5: Multi-scale SGM score maps under three types of text prompts for OCT images. All score maps share a unified global normalization range for fair comparison. Columns show noisy input and score maps at three scales; rows correspond to the correct prompt "speckle in OCT", semantically related prompt "speckle in ultrasound", and unrelated prompt "noise in low-dose CT". The related prompt produces similar spatial activation, while the unrelated prompt generates distinct noisy responses.

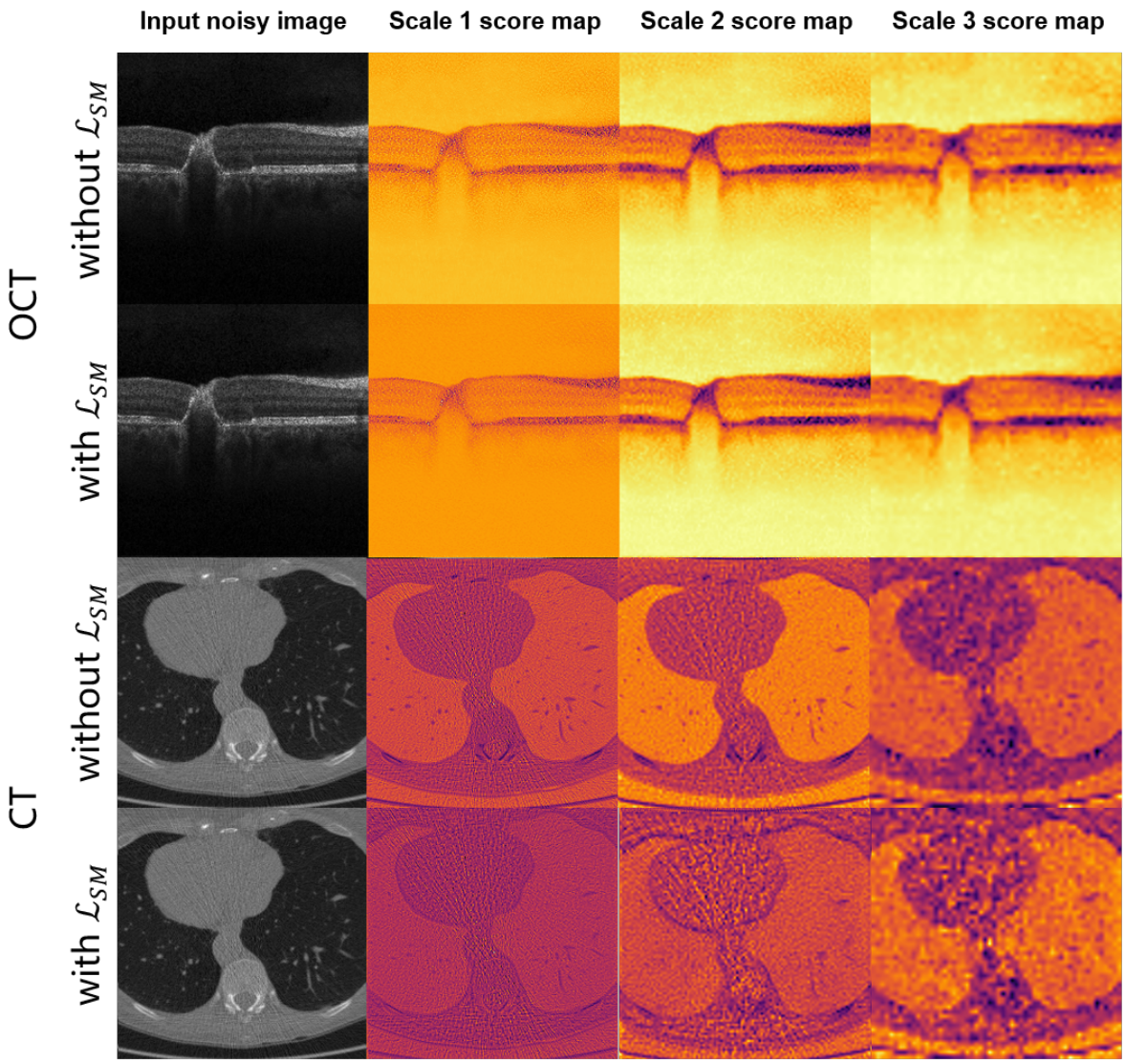


Figure S6: Comparison of SGM score maps trained with and without $\mathcal{L}_{\text{SM}}$ (OCT and CT samples shown). Rows: with $\mathcal{L}_{\text{SM}}$ loss, without $\mathcal{L}_{\text{SM}}$ loss. Columns: noisy input, scale-1, scale-2, scale-3 score maps.